

HASSI MESSAOU, Algeria

WMO: 605810

Lat: 31.673N

Lon: 6.140E

Elev: 463

StdP: 14.45

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	35.8	38.7	19.0	14.5	61.2	21.7	16.6	60.5	31.0	62.7	27.2	62.2	1.7	200	0.520

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.4	115.2	72.0	113.1	71.5	110.8	70.7	75.4	102.5	73.8	102.4	72.7	102.0	11.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.9	104.2	86.1	65.5	95.9	84.8	63.2	88.2	84.5	39.0	101.3	37.5	101.4	36.4	101.9	90.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.7	24.0	20.7	DB	31.4	118.8	2.0	1.7	30.0	120.0	28.8	121.0	27.7	122.0	26.3	123.2
			WB	28.9	80.3	2.4	3.7	27.1	83.0	25.7	85.2	24.3	87.3	22.6	90.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	74.7	53.0	57.0	65.0	73.2	82.6	90.9	96.3	95.0	87.6	76.6	63.1	54.9
	DBStd	16.33	4.32	5.94	6.94	7.11	7.02	6.15	4.78	4.26	6.32	6.41	5.83	4.96
	HDD50	36	18	7	0	0	0	0	0	0	0	0	0	11
	HDD65	1125	371	234	87	10	0	0	0	0	0	2	104	317
	CDD50	9055	112	203	466	696	1010	1227	1436	1394	1129	824	395	163
	CDD65	4669	0	10	88	256	546	777	971	929	679	361	48	4
	CDH74	73508	10	174	1126	3229	7868	12952	17301	16058	10182	4084	485	39
	CDH80	49009	0	41	411	1601	4784	9000	12884	11644	6519	2003	116	6
Wind	WSAvg	8.3	6.5	8.1	9.2	10.3	10.5	10.0	8.5	8.2	8.6	7.1	6.3	6.4
Precipitation	PrecAvg	1.6	0.3	0.1	0.3	0.2	0.2	0.0	0.0	0.0	0.2	0.1	0.2	0.2
	PrecMax	4.6	2.3	1.1	1.9	1.8	0.9	0.1	0.0	0.4	1.3	0.6	1.3	0.9
	PrecMin	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	0.9	0.5	0.2	0.4	0.4	0.2	0.0	0.0	0.1	0.3	0.2	0.3	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.1	84.8	93.5	102.5	111.2	116.3	118.8	116.9	112.8	102.6	87.6	78.9
		MCWB	54.9	57.8	61.6	65.7	69.5	71.4	72.5	72.7	70.6	68.7	61.8	58.6
	2%	DB	71.2	78.9	89.2	97.6	105.9	113.1	116.6	114.4	108.6	98.4	83.3	72.9
		MCWB	53.9	56.1	60.3	64.3	67.8	70.3	72.5	72.0	70.3	66.5	60.7	55.6
	5%	DB	68.0	74.8	84.4	93.5	102.3	110.1	114.3	111.4	105.7	94.1	79.5	69.5
		MCWB	52.8	54.6	58.5	62.8	66.9	69.6	72.0	71.2	69.7	66.1	59.4	54.3
	10%	DB	64.8	70.1	80.5	89.4	98.6	106.7	111.3	109.0	102.2	90.0	75.6	66.4
		MCWB	51.1	52.5	57.1	61.6	66.0	68.7	71.2	70.9	69.4	65.3	57.8	52.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.8	61.6	65.4	69.1	73.8	74.6	78.6	77.8	75.3	72.3	66.3	61.0
		MCDB	69.7	77.7	81.3	92.5	97.4	102.3	105.9	104.0	95.6	91.4	80.4	73.1
	2%	WB	55.5	57.7	62.1	66.3	70.6	72.2	75.2	75.4	73.5	70.0	63.4	57.5
		MCDB	67.4	75.7	82.2	89.9	97.2	103.4	106.2	102.6	96.2	87.9	77.6	68.8
	5%	WB	53.7	55.2	59.9	64.5	68.6	70.9	73.5	73.8	72.3	68.5	61.1	55.6
		MCDB	65.5	71.5	81.2	87.7	95.9	102.6	106.3	102.6	95.7	86.0	75.5	66.9
	10%	WB	51.9	53.2	58.0	62.7	66.9	69.7	72.0	72.6	71.1	67.1	59.1	54.0
		MCDB	63.4	68.4	77.9	85.3	94.2	101.7	106.0	102.3	94.9	84.8	73.2	64.9
Mean Daily Temperature Range	5% DB	MDBR	23.9	24.3	25.1	25.1	25.2	25.8	26.4	25.2	24.0	23.5	23.4	22.6
		MCDBR	27.7	29.2	30.0	30.4	28.9	29.6	29.3	28.3	28.3	27.6	26.4	25.7
		MCWBR	14.8	14.1	13.1	12.0	10.9	10.6	10.4	9.5	9.6	10.4	11.5	13.6
	5% WB	MCDBR	25.3	26.2	27.0	26.5	25.9	26.2	26.4	24.5	23.8	22.8	23.3	22.9
		MCWBR	14.4	13.6	12.9	12.1	11.4	10.6	10.7	9.9	9.1	9.8	11.6	13.0
Clear-Sky Solar Irradiance	taub	0.371	0.398	0.506	0.550	0.605	0.601	0.619	0.592	0.594	0.519	0.426	0.383	
	taud	2.093	2.008	1.731	1.663	1.574	1.593	1.577	1.636	1.641	1.774	1.956	2.062	
	Ebn at Noon	262	267	245	241	229	229	224	227	219	225	239	249	
	Edh at Noon	41	50	71	79	87	85	86	80	76	62	47	41	
All-Sky Solar Radiation	RadAvg	1185	1483	1815	2140	2299	2453	2443	2242	1885	1534	1222	1054	
	RadStd	46	63	71	82	79	72	34	47	61	60	44	49	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page